Year 1 AS Statistics and Mechanics

Chapter 2

Measures of location and spread

Practice Booklet

Contents

2.1 Measures of central tendency ..... *book p. 8*
2.2 Other measures of location ..... *book p. 9*
2.3 Measures of spread ..... *book p. 10*
2.4 Variance and standard deviation ..... *book p. 12*
2.5 Coding ..... *book p. 13*
Problem solving: Set A ..... *book p. 14*
Problem solving: Set B ..... *book p. 15*
Selected exam question bank questions .....
Answers .....

2.1 Measures of central tendency

Practice Book, printed page 8.

1. The masses, in grams, of 9 pieces of cheese are:

550 450 300 400 450 500 450 300 600

Calculate:

a. the mode

b. the mean

c. the median.

Hint

To find the median, put the data in ascending order.

2. The table shows the frequency distribution for the numbers of eggs in the nests of a group of blackbirds.

Number of eggs	2	3	4	5	6
Frequency	6	11	13	7	2

Find:

a. the mode

b. the mean

c. the median.

Hint

To calculate the mean, use the formula $\bar{x} = \dfrac{\sum xf}{\sum f}$

3. The table shows the times taken by a group of students to walk to school.

Time, t (mins)	$10 \leq t < 15$	$15 \leq t < 20$	$20 \leq t < 25$	$25 \leq t < 30$
Frequency	5	14	7	2

a. Write down the modal class.

b. Calculate an estimate for the mean time taken to walk to school.

Hint

Use the midpoint of each class interval.

4. E/P In class 12B, there are 13 girls and 11 boys. The mean height of the girls is 1.60 m and the mean height of the boys is 1.71 m. Calculate the mean height of all 24 pupils. (3 marks)

5. E Bill is shelling peas. He records the number of peas inside each pod and uses a frequency table to show his results.

Number of peas	4	5	6	7	8
Frequency	12	15	18	11	7

a. Write down the median for this data. (1 mark)

b. Calculate the mean number of peas per pod. (1 mark)

He opens one more pod and discovers it contains 7 peas.

c. Write down the effect that including this piece of data will have on the mean. (1 mark)

6. E A convenience store manager records the amounts of time customers spend in the store.

Time, t (mins)	$2 \leq t < 4$	$4 \leq t < 6$	$6 \leq t < 8$	$8 \leq t < 10$
Frequency	7	9	12	6

a. Write down the modal class. (1 mark)

b. Calculate an estimate for the mean amount of time spent in the store. (1 mark)

c. Explain why your answer to part **b** is an estimate. (1 mark)

7. E/P Using data compiled from the large data set, the table shows the mean daily temperatures in Jacksonville during May 2015.

Temp, t (°C)	$16 \leq t < 18$	$18 \leq t < 20$	$20 \leq t < 22$	$22 \leq t < 24$	$24 \leq t < 26$	$26 \leq t < 28$
Frequency	1	3	2	9	13	3

© Crown Copyright Met Office

a. Write down the modal class. (1 mark)

b. Calculate an estimate for the mean daily temperature. (1 mark)

2.2 Other measures of location

Practice Book, printed page 9.

1. The masses, in grams, of seven gold ingots are:

324 313 315 372 377 312 369

Write down:

a. the lower quartile

b. the upper quartile.

Hint

Put the data in ascending order first.

2. The table shows the frequency distribution of the numbers of siblings in a group of families.

Number of siblings	1	2	3	4	5
Frequency	7	12	15	11	6

Write down:

a. the median

b. the upper quartile

c. the lower quartile.

Hint

Find half of n, a quarter of n and three quarters of n. If these are not whole numbers, round up and pick this data point.

3. The table shows the distances travelled by a group of students to get to school.

Distance, x (km)	$1 \leq x < 3$	$3 \leq x < 5$	$5 \leq x < 7$	$7 \leq x < 9$
Frequency	6	8	5	3

a. Use linear interpolation to estimate the median distance travelled.

b. Use linear interpolation to estimate the upper quartile.

Hint

Work out which classes contain the median and upper quartile. Then use proportion to estimate out how far through each class the measure of location will be found.

4. E/P Jonas records the number of books on each of 14 shelves in his library:

13 14 12 18 20 14 17
15 15 21 20 15 17 20

a. Calculate the median number of books. (2 marks)

b. Calculate the lower quartile. (2 marks)

5. E/P The masses, in kg, of some students are recorded in the table.

Mass, m (kg)	$40 \leq m < 45$	$45 \leq m < 50$	$50 \leq m < 55$	$55 \leq m < 60$
Frequency	7	10	14	5

a. Use linear interpolation to find an estimate for the median mass. (2 marks)

b. Explain why your answer to part **a** is an estimate. (1 mark)

6. E/P The lengths, in cm, of a group of slowworms are recorded in the table.

Length, x (cm)	$25 \leq x < 30$	$30 \leq x < 35$	$35 \leq x < 40$	$40 \leq x$
Frequency	25	28	17	12

a. Use linear interpolation to estimate:

i. the median length

ii. the 70th percentile.

(3 marks)

b. Explain why it is not possible to estimate the 90th percentile in this way. (1 mark)

7. E/P The lengths of train delays at a station on a particular day are recorded in the table.

Time, t (mins)	$1 \leq t < 3$	$3 \leq t < 5$	$5 \leq t < 10$	$10 \leq t < 20$
Frequency	6	13	18	4

a. Estimate the 60th percentile. (2 marks)

b. State one assumption you have made in your answer to part **a**. (1 mark)

The station advertises that fewer than 20% of trains are delayed by more than 8 minutes.

c. By calculating a suitable percentile, comment on the validity of this claim. (3 marks)

2.3 Measures of spread

Practice Book, printed page 10.

1. The lengths, in cm, of 9 dachshunds are:

45 48 52 60 51 53 47 49 50

a. Find the range for this data.

b. Calculate the interquartile range.

Hint

Put the data in ascending order first.
Interquartile range $= Q_3 - Q_1$

2. The table shows the number of pets owned by a class of students.

Number of pets	0	1	2	3	4
Frequency	2	6	8	13	1

Calculate:

a. the range

b. the interquartile range.

Hint

For part **b**, work out Q_1 and Q_3 first.

3. The table shows the lengths of time, in minutes, it takes 31 students to complete a test.

Time, t (mins)	$40 \leq t < 42$	$42 \leq t < 44$	$44 \leq t < 46$	$46 \leq t < 48$
Frequency	2	15	11	3

a. Write down an estimate for the range.

b. Calculate an estimate for the interquartile range.

Hint

For part **b**, use linear interpolation to find Q_1 and Q_3.

4. E/P From the large data set, the daily mean windspeeds (knots) in Beijing for the first 10 days of May 2015 are:

5.5 3.8 7.5 8.0 5.5 4.0 5.3 5.0 3.3 3.5

a. Calculate the median and interquartile range. (2 marks)

The median daily mean windspeed in Beijing for the first 10 days of June 2015 was 4.1 knots and the interquartile range was 3 knots.

b. Compare the data for May with the data for June. (2 marks)

5. E/P The masses, in grams, of a group of 4-week-old kittens are recorded in the table.

Mass, m (g)	$400 \leq m < 425$	$425 \leq m < 450$	$450 \leq m < 475$	$475 \leq m < 500$
Frequency	18	25	28	13

a. Estimate the range for this data. (1 mark)

b. Explain why the actual range of the data will differ from your answer to part **a**. (1 mark)

c. Calculate the 30% to 70% interpercentile range. (3 marks)

d. Estimate the number of kittens whose masses fall within the range you calculated in part **c**. (2 marks)

6. E/P The lengths of time customers are kept on hold at a company call centre are recorded in the table.

Time, t (mins)	$0 \leq t < 4$	$4 \leq t < 8$	$8 \leq t < 12$	$12 \leq t < 16$	$16 \leq t < 20$
Frequency	6	18	27	32	17

The company claims that 90% of customers wait between 2 and 16 minutes for their call to be answered. By calculating suitable percentiles, comment on the validity of the company's claim. (3 marks)

2.4 Variance and standard deviation

Practice Book, printed page 12.

1. The scores, marked out of 10, achieved by 6 students in a quiz are:

5 8 6 7 8 4

Find:

a. the variance

b. the standard deviation.

Hint

Use the formula,

$$\text{variance} = \frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2$$

The standard deviation is the square root of variance.

2. The table shows some grouped data.

x	0	1	2	3	4
f	4	8	9	17	6

Calculate:

a. $\sum fx^2$

b. the standard deviation.

Hint

For part **b**, use the formula for standard deviation of grouped data

$$\sigma = \sqrt{\frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f}\right)^2}$$

Alternatively, enter the values into your calculator.

3. The table shows the masses of children in a year 9 class.

Mass, m (kg)	**Frequency**
$35 \leq m < 40$	5
$40 \leq m < 45$	7
$45 \leq m < 50$	8
$50 \leq m < 55$	6
$55 \leq m < 60$	4

Calculate an estimate for:

a. the variance

b. the standard deviation.

Hint

Use the midpoint of each class either in the formula given in the hint above, or as column 1 in a table on your calculator.

4. E/P A keen gardener has many rose bushes. One day the gardener counts the number of ladybirds on each bush and records the data in a table.

Number of ladybirds	**Frequency**
5	11
6	15
7	23
8	16

a. Use your calculator to work out the mean and standard deviation of the number of ladybirds on a rose bush. (2 marks)

The gardener repeats the count of ladybirds two weeks later and finds the mean number to be 7.1 and the standard deviation to be 0.94.

b. Compare the data on the two days. (2 marks)

5. E/P Jez is a postman. For each day in one working month he records the time it takes him, in hours, to complete his round.

Time, t (hours)	$4 \leq t < 4.5$	$4.5 \leq t < 5$	$5 \leq t < 5.5$	$5.5 \leq t < 6$	$6 \leq t < 6.5$
Frequency	5	7	8	3	1

a. Use your calculator to estimate the mean and standard deviation of the data. (2 marks)

Jez claims that he completes 80% of his rounds within one standard deviation of the mean time.

b. Comment on Jez's claim. (2 marks)

6. E/P A set of 6 numbers has mean 5 and variance 7.
A second set of 5 numbers has mean 5.5 and variance 6.5.
Find the mean and variance of the combined set of 11 numbers. (4 marks)

7. E From the large data set for Camborne in August 2015, the following summary statistics are collected for the daily mean temperature.

$$\sum x = 477.4 \qquad \sum x^2 = 7374.7$$

Find the standard deviation of the daily mean temperatures. (2 marks)

2.5 Coding

Practice Book, printed page 13.

1. A set of data values, d, is shown below:

$$60 \quad 70 \quad 80 \quad 40 \quad 50 \quad 80 \quad 70$$

a. Code the data using the coding $x = \dfrac{d}{10}$

b. Calculate the mean of the coded values, $\bar{x}$.

c. Use your answer to part **b** to calculate $\bar{d}$.

Hint

If $x = \dfrac{d}{10}$ then $\bar{x} = \dfrac{\bar{d}}{10}$

2. A set of data is coded using $y = 2x - 1$. It is found that $\bar{y} = 15$ and $\sigma_y = 6$.
Find:

a. the mean of the original data, $\bar{x}$

b. the standard deviation of the original data, σ_x

c. the variance of the original data.

Hint

When data is coded using $y = px + q$, the standard deviation and variance are only affected by the value of p.

3. E The prices of packs of apples, x pence, in ten supermarkets are recorded.
The data is coded using $y = \dfrac{x - 20}{100}$ and the following summary statistics are obtained:

$$\sum y = 20 \qquad \sum y^2 = 48$$

Find:

a. the mean price of a pack of apples (2 marks)

b. the standard deviation of the prices of packs of apples. (2 marks)

4. E A meteorologist records the daily mean pressures, p hPa, at Heathrow on the first 10 days of May 2015.
The data is coded using $x = \dfrac{p - 1000}{2}$. The summary statistics are:

$$\sum x = 43.5 \qquad \sum x^2 = 365.25$$

Find:

a. the mean of the daily mean pressures (2 marks)

b. the variance of the daily mean pressures. (2 marks)

5. E/P Yvonne is investigating the salaries of top executives in the banking industry.

a. Given that a typical salary is in excess of £100 000, explain why it might be sensible for Yvonne to code her raw data. (1 mark)

She records the salaries of 8 people and uses the coding $y = ax + b$ where x is the original data and y is the coded data. Her coded summary statistics are:

$$\sum y = 104 \qquad \sum y^2 = 1634$$

b. Given that the mean salary in the sample is £165 000 and the variance is 881 250 000, find the values of a and b. (3 marks)

Problem solving: Set A

Bronze

The number of tracks on each of 11 albums is shown below:

12 13 12 8 9 11 12 10 9 13 14

Calculate the interquartile range for this data. (2 marks)

Silver

In a survey, a random sample of 140 people were asked how long they spent commuting to work every week, to the nearest hour. The results are recorded in the table.

Number of hours	Frequency
0–2	19
3–4	47
5–6	57
7–10	17

Use linear interpolation to find the interquartile range for this data. (3 marks)

Gold

The table shows the scores achieved by 160 students in a mock examination.

Score, x%	Frequency
$0 \leq x < 20$	12
$20 \leq x < 40$	31
$40 \leq x < 60$	58
$60 \leq x < 80$	44
$80 \leq x \leq 100$	15

a. Use linear interpolation to find the 10th to 90th interpercentile range of this data. (3 marks)

The next year, 200 students will sit a similar mock examination. The school wants to set a pass mark such that 175 of these students will pass.

b. Based on this year's mock examination scores, suggest a suitable pass mark. (2 marks)

c. State one assumption you have made in your calculation in part **b**. (1 mark)

Problem solving: Set B

Bronze

Stevo is a keen fisherman. Over the course of a year, he records the amount of time he spends fishing on each day he goes out. His results are summarised in the table.

Time, t (hours)	$2 \leq t < 3$	$3 \leq t < 4.5$	$4.5 \leq t < 5$	$5 \leq t < 7$	$7 \leq t < 10$
Frequency	18	40	51	36	62

a. Estimate the mean and standard deviation for this data. (2 marks)

b. Explain why your answers to part **a** are estimates. (1 mark)

Silver

Data is collected on the masses, m kg, of 140 Humboldt penguins.

Mass, m (kg)	$3 \leq m < 3.5$	$3.5 \leq m < k$	$k \leq m < 4.0$	$4.0 \leq m < 4.5$	$4.5 \leq m < 6$
Frequency	17	21	33	54	15

a. Given that an estimate for the mean mass of the penguins is 4.09 kg, find the value of k. (3 marks)

b. Calculate an estimate for the standard deviation. (1 mark)

c. Explain why your answer to part **b** is an estimate. (1 mark)

Gold

The salaries of 50 employees in an IT company are recorded in the table.

Salary, £'000	Frequency
$10 \leq x < 12$	2
$12 \leq x < 15$	5
$15 \leq x < 20$	12
$20 \leq x < 25$	13
$25 \leq x < 30$	7
$30 \leq x < 50$	8
$50 \leq x$	3

The human resources manager wants to calculate an estimate for the mean salary.

a. Identify one problem that the manager will have in calculating an estimate using the data given. (1 mark)

b. Suggest a possible assumption that the manager could make to rectify this problem. (1 mark)

c. Using your assumption in part **b**, calculate an estimate for the mean salary. (2 marks)

In reality, the mean salary was found to be £31 000.

d. Comment on your assumption from part **b** in light of this information, and suggest a reason for any discrepancy. (2 marks)

Now try this → Exam question bank Q6, Q13, Q15, Q24, Q30

Selected exam question bank questions

Question 6

A tomato farmer is investigating the ripeness and consistency of his crop. To test the ripeness of a tomato he picks and squeezes it.

a. Explain why it would not be sensible for the farmer to use a census to test for ripeness. (1 mark)

He measures the consistency of the crop by weighing the tomatoes. He records the masses, to the nearest gram, of a random sample of tomatoes.

Mass (g)	110–119	120–129	130–139	140–149	150–159
Frequency	7	8	x	10	4

b. Given that an estimate for the mean mass of a tomato from the sample is 133.5 g, find x. (3 marks)

c. Calculate an estimate for the standard deviation of the masses. (1 mark)

A second farmer claims that her crop is more consistent in mass. She calculates the standard deviation of a random sample of 5 tomatoes and finds it to be 10.1 g.

d. Comment on the validity of this claim. (2 marks)

Question 13

Naomi is interested in the amount of time, to the nearest hour, that people spend online during a typical week. By using opportunity sampling, she collects a sample of 25 people from shoppers on the high street.

a. Explain what is meant by opportunity sampling. (1 mark)

b. State, with a reason, whether the data that Naomi has collected is discrete or continuous. (1 mark)

Naomi's results are recorded in the table.

Number of hours	**Frequency**
5–9	1
10–12	3
13–16	7
17–25	12
26–40	2

c. Use linear interpolation to find an estimate of the interquartile range for this data. (3 marks)

Question 15

Niamh is investigating the daily mean windspeed, in knots, in Perth in 2015. She wants to take a random sample of size 30 from all 194 dates in the large data set.

a. Explain briefly how Niamh could take her sample. (2 marks)

The data that Niamh collects is summarised in the table.

Windspeed, w (kn)	$4 \leq w < 6$	$6 \leq w < 8$	$8 \leq w < 10$	$10 \leq w < 12$
Frequency	5	13	8	4

© Crown Copyright Met Office

b. Use linear interpolation to estimate the median daily mean windspeed. (2 marks)

c. Use your calculator to find estimates for the mean and standard deviation of the daily mean windspeeds. (2 marks)

Question 24

Hafsah is investigating the daily mean pressure, x hPa, in Jacksonville in 1987. She takes a random sample of size 10 from the large data set.

Hafsah codes the data using $y = \dfrac{x - 1010}{4}$

Summary statistics for the coded data are:

$$\sum y = 21 \qquad \sum y^2 = 44.625$$

a. Calculate the mean and variance of the coded data. (3 marks)

b. Use your answer to part **a** to find the mean and standard deviation of the data in Hafsah's sample. (2 marks)

Mishel says that Hafsah's results are unreliable.

c. Suggest a way in which Hafsah can improve the reliability of her results. (1 mark)

Question 30

A teacher is investigating the times taken, t minutes, for her class to complete a maths quiz. She produces a table of coded times, x minutes, for a random sample of 32 students.

Coded time (x minutes)	**Frequency (f)**	**Coded time midpoint (y minutes)**
$0 \leq x < 2$	3	1
$2 \leq x < 5$	11	3.5
$5 \leq x < 10$	7	7.5
$10 \leq x < 15$	8	12.5
$15 \leq x < 20$	3	17.5

You are given these summary statistics:

$$\sum fy = 246.5 \qquad \sum fy^2 = 2700.25$$

a. Use linear interpolation to estimate the median of the coded times. (2 marks)

b. Estimate the standard deviation of the coded times. (2 marks)

The head of department is told by the teacher that she subtracted 10 from each of the times and then divided by 2 to calculate the coded times.

c. Calculate an estimate for the median and standard deviation of the actual times. (2 marks)

In a different quiz, students can qualify for a national competition. There are a maximum of 10 places available, and students must complete the quiz in 16 minutes or less to qualify. 45 Students take part in the quiz.

d. Will the teacher fill all of the places available? Give a reason for your answer. (2 marks)

Answers

2.1 Measures of central tendency

1.

a. 450 g

b. 444 g (3 s.f.)

c. 450 g

2.

a. 4

b. 3.69 (3 s.f.)

c. 4

3.

a. $15 \leq t < 20$

b. 18.6 minutes (3 s.f.)

4. 1.65 m (3 s.f.)

5.

a. 6

b. 5.78 (3 s.f.)

c. It will increase it.

6.

a. $6 \leq t < 8$

b. 6 minutes

c. The midpoint has been used to estimate the time each customer in each class interval has spent in the store.

7.

a. $24 \leq t < 26$

b. 23.5°C (3 s.f.)

2.2 Other measures of location

1.

a. 313 g

b. 372 g

2.

a. 3

b. 4

c. 2

3.

a. 4.25 km

b. 6 km

4.

a. 16

b. 14

5.

a. 50.4 kg (3 s.f.)

b. It is assumed that the frequency in each class is evenly distributed throughout the interval.

6.

a.

i. 32.9 cm (3 s.f.)

ii. 36.3 cm (3 s.f.)

b. The 90th percentile is in the last class which has no upper class boundary.

7.

a. 6.56 minutes (3 s.f.)

b. The data is evenly distributed within each class.

c. The 80th percentile is 8.83 mins (3 s.f.) so the advertisement is incorrect. In fact, 20% of trains are delayed by more than 8.83 minutes.

2.3 Measures of spread

1.

a. 15 cm

b. 5 cm

2.

a. 4

b. 2

3.

a. 8 minutes

b. 2.37 minutes (3 s.f.)

4.

a. Median 5.15 knots; interquartile range 1.7 knots

b. The median daily mean windspeed was higher for the 10 days in May and there was less variability in windspeeds.

5.

a. 100 g

b. It will be less than this as it is unlikely that there is a value of 400 g and there cannot be a value of 500 g.

c. 31.9 g (3 s.f.)

d. 33 kittens

6. The 5% and 95% percentiles are 3.33 and 18.8 so the company is understating the wait times.

2.4 Variance and standard deviation

1.

a. 2.22 (3 s.f.)

b. 1.49 (3 s.f.)

2.

a. 293

b. 1.18 (3 s.f.)

3.

a. 40.6 kg (3 s.f.)

b. 6.37 kg (3 s.f.)

4.

a. Mean 6.68 ladybirds; standard deviation 1.02 ladybirds

b. The mean number of ladybirds is higher on the second day, but there was less variability.

5.

a. Mean 5 hours; standard deviation 0.540 hours (3 s.f.)

b. One standard deviation from the mean gives $[4.46, 5.54]$, which gives a frequency of 15 inside this interval. This is only 62.5% of the rounds, so Jez is wrong.

6. Mean 5.23 (3 s.f.); variance 6.83 (3 s.f.)

7. 0.856°C (3 s.f.)

2.5 Coding

1.

a. 6 7 8 4 5 8 7

b. 6.43 (3 s.f.)

c. 64.3 (3 s.f.)

2.

a. 8

b. 3

c. 9

3.

a. 220 p or £2.20

b. 89.4 p

4.

a. 1009 hPa (4 s.f.)

b. 70.41 hPa^2

5.

a. Coding makes large numbers easier to manage.

b. $a = \frac{1}{5000}$ and $b = -20$

Problem solving: Set A

Bronze 4

Silver 2.69 hours (3 s.f.)

Gold

a. 57.0 (3 s.f.)
b. 25%
c. That the scores are evenly distributed throughout the class, or that next year's scores will be similarly distributed to this year's scores.

Problem solving: Set B

Bronze

a. Mean 5.70 hours (3 s.f.); standard deviation 2.05 hours (3 s.f.)
b. It is assumed all of the data points in each class are equal to the midpoint of the class interval.

Silver

a. 3.94 (3 s.f.)
b. 0.514 kg (3 s.f.)
c. It is assumed all of the data points in each class are equal to the midpoint of the class interval.

Gold

a. The last class is open-ended.
b. The manager could assume a maximum value for the class, such as 100.
c. Using 100 as the upper class boundary of the last class gives a mean of £26 590.
d. There could be a single very high value (such as the salary of the CEO) that makes the assumption in part **b** invalid.

Selected exam question bank answers

Question 6

a. The testing is likely to damage the tomatoes.
b. 11
c. 12.4 g (3 s.f.)
d. The standard deviation of the second farmer's crop is lower so there is evidence to suggest greater consistency, but her sample size is also very small and therefore the results are likely to be unreliable.

Question 13

a. The person doing the survey asks the first n people they meet at a specific time and/or place.
b. Continuous. Time can take any value in the range. Just because it is rounded to the nearest hour does not negate this fact.
c. 8.53 hours (3 s.f.)

Question 15

a. Have an ordered list of dates, and generate 30 random numbers between 1 and 194.
b. 7.54 kn (3 s.f.)
c. Mean 7.73 kn (3 s.f.); standard deviation 1.82 kn (3 s.f.)

Question 24

a. Mean 2.1; variance 0.0525

b. Mean 1018.4 hPa; standard deviation 0.917 hPa (3 s.f.)

c. Increase the size of her sample.

Question 30

a. 6.43 (3 s.f.)

b. 5.00 (3 s.f.)

c. Median 22.86 minutes; standard deviation 10 minutes (using rounded values from **a** and **b**)

d. No. The percentage of students in the sample finishing the puzzle in less than 16 minutes is 20.8%. So assuming the same distribution of times in the selection round, she will only get 9 students ($45 \times 20.8 = 9.36$).